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time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, strings | https://codeforces.com/problemset/problem/2180/B

You have an array a of n strings $a_1, a_2, \ldots, a_n$, each consisting of lowercase English letters, and an empty string s.

In the i-th ($1 \le i \le n$) step, you should do one of the following:

- add a_i to the beginning of s, or
- add a_i to the end of s.

For example, if before the i-th step s = mathtt*aba* and a_i = mathtt*bba*, after the i-th step, s will be equal to mathtt*ababba* or mathtt*bbaaba*.

What's the lexicographically smallest string s you can reach after n steps?

A string a is lexicographically smaller than a string b of the same length, if and only if the following holds:

- in the first position where a and b differ, the string a has a letter that appears earlier in the alphabet than the corresponding letter in b.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 500$). The description of the test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 1000$) — size of array a. The next line contains n strings $a_1, a_2, \ldots, a_n$ ($1 \le |a_i| \le 4000$), each consisting of lowercase English letters.

It is guaranteed that the sum of n over all test cases does not exceed 1000, and **the total length of all strings in the input** (over all test cases) does not exceed 4000.

Output

For each test case, print the lexicographically minimum string s you can reach after n steps.

Example

Input
3 4 amir rima amin nima 1 codeforces 3 a ab abc

Output
aminamirrimanima codeforces aababc

Note

In the first test case, one possible way to construct the lexicographically minimum string s is as follows:

1. After the first step, s = mathtt*amir* regardless of whether we add it to the beginning or the end, since s was initially empty.
2. In the second step, we add a_2 = mathtt*rima* to the end of s. Now s = mathtt*amirrima*.
3. In the third step, we add a_3 = mathtt*amin* to the beginning of s. Now s = mathtt*aminamirrima*.
4. In the last step, we add a_4 = mathtt*nima* to the end of s. Therefore, the final value of s is mathtt*aminamirrimanima*.

It can be proven that this resulting string is indeed the lexicographically smallest string obtainable after all steps.